

Sensor Re-allocation Mapping

Derived from the QSYS instrumentation-location study (LB / LBI) and the QM instrumentation & controls brainstorming. The goal is to monitor the Piezo (PZ) tuner thermal gradient and the MAG / coupler thermalisation points with the correct cryogenic sensor technology.

1. Sensor technologies

Sensor type	Technology	Best for
TT-CX	Cernox® resistance thermometer	Coldest points (≤ 4 K range), high sensitivity at low T
TT-PT100	Pt-100 RTD	Warmest points (> 40 K range), stable at higher T

2. Primary re-allocations (PZ — Piezo tuner)

Original tag	New tag	Sensor type	Position	In source geometry
TT535	PZ535	TT-CX	Coldest part of the Piezo (PZ)	✓ yes — re-tagged & ring-highlighted on drawing
TT525	PZ525	TT-PT100	Warmest part of the Piezo (PZ)	⚠ design re-allocation — tag not present in current source extract; applied automatically if/when present

On the drawings, a re-allocated sensor is shown as a normal ISA bubble with a dashed **red highlight ring** and a red sensor-type label (TT-CX / TT-PT100).

3. MAG & coupler-port redistribution

TT-CX and TT-PT100 sensors are redistributed to monitor the MAG component and the coupler thermalisation ports:

Sensor type	Location	Purpose
TT-CX	MAG cold port	Cold-end temperature of the MAG
TT-PT100	MAG warm port	Warm-end temperature of the MAG
TT-CX	Coupler port (cold)	Coupler thermalisation, cold side
TT-PT100	Coupler port (warm)	Coupler thermalisation, warm side

4. PZ monitoring strategy (summary)

- **TT-CX** → **coldest part of the Piezo** (maximum sensitivity in the 2–4 K band).
- **TT-PT100** → **warmest part of the Piezo** (stable readout in the 40–300 K band).
- The pair brackets the PZ thermal gradient so the control system can verify that the actuator stays within its allowable temperature window.

5. Where to see it

- Re-allocated PZ sensors appear on the **Instrumentation sheets** (Sheet 2) of both QCELL and RFCELL, on **Layer 9 — Instruments** (bubble) with the red highlight, and the sensor-type callout on **Layer 12 — Tags**.
- The machine-readable record is in `_build_meta.json` under `realloc / mag_coupler`.